

ABDULRAHMAN ALSAADI

MECHATRONICS ENGINEER | MECHANICAL DESIGN

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PROFESSIONAL SUMMARY

CSWP-certified Mechatronics Engineer specializing in mechanical design, product development, and electromechanical systems. Hands-on experience designing, analyzing, manufacturing, assembling, and physically validating robotic systems using SolidWorks, FEA, DFM, dimensional tolerancing, bearing fits, topology optimization, torque analysis, and additive manufacturing. Experienced in taking designs from CAD and engineering analysis through prototyping, assembly, testing, and iterative redesign.

TECHNICAL SKILLS

Mechanical Design & Analysis: SolidWorks, FEA, ANSYS, DFM, Manufacturing Drawings, Dimensional Tolerancing, Bearing Fits, Topology Optimization, Mechanism Design, 3D Printing.

Robotics & Electromechanical Systems: Robotic Mechanisms, Belt/Gear Transmissions, Stepper/Servo/BLDC Motors, Raspberry Pi 5, ESP32, Arduino, Cube Orange

Simulation & Software: ROS 2 Humble, Gazebo, C++, Python, OpenCV, ArduPilot SITL, MAVLink/pymavlink, Linux (Ubuntu), Git

SELECTED ENGINEERING PROJECTS

WD DRONE TEAM - TEKNOFEST 2026 Coemption, Rotary Wing Category

Karabük, Türkiye

Founding Team Member | Simulation & Systems Integration

Feb 2026 - Jul 2026

- Contributed to the mechanical design, prototyping, assembly, and physical validation of the UAV, including 3D-printed components, dimensional tolerances, and iterative redesign based on fit and assembly requirements.
- Performed SolidWorks Simulation structural FEA on the UAV's 5 mm carbon-fiber arm plates and 3 mm central-body plates under an 8 kg static load, obtaining 24.9 MPa maximum von Mises stress and a calculated 31.1 FoS using supplier material data.
- Designed and validated the UAV simulation environment in Gazebo/ArduPilot SITL for autonomous waypoint survey, target approach, payload delivery, and RTL mission testing.
- Integrated Raspberry Pi 5, Raspberry Pi Camera Module 3, and Cube Orange through MAVLink/pymavlink; developed OpenCV-based target detection at approximately 30 FPS.
- Physically tested autonomous flight, target detection, and payload release across five flight-test sessions at approximately 3–10 m altitude.

SCARA ROBOTIC ARM - GRADUATION PROJECT

Karabük, Türkiye

Mechatronics & Robotics Engineer

Apr 2026 - Jun 2026

- Independently designed, manufactured, assembled, and tested a 3-DOF SCARA robotic arm with 492.5 mm reach and 100 g payload, including custom components, subassemblies, bearing housings, transmission components, and motor interfaces in SolidWorks.
- Applied DFM, dimensional tolerancing, and bearing-fit principles; redesigned components after identifying physical fit and assembly issues during prototyping.
- Produced engineering drawings, BOM, and manufacturing-ready CAD for the complete mechanical system.
- Performed topology optimization on the arm links, reducing design mass by approximately 80% while retaining critical mounting and load-bearing regions; manufactured and physically tested the optimized components.
- Designed and integrated a two-stage GT2 belt/gear transmission, NEMA17 motor system, bearings, and servo gripper; performed joint torque calculations and validated reach, payload handling, motion, and mechanical assembly.
- Manufactured components using FDM 3D printing with PLA+ and performed post-processing including sanding, drilling, reaming, heat-set inserts, and tapping.

EXPERIENCE

SMART METHODS

Jeddah, Saudi Arabia

Full Stack Robotics Engineering Intern

Jul 2025 - Aug 2025

- Completed 13 engineer-evaluated robotics projects spanning mechanical design, analysis, embedded systems, and robotic-system development.
- Redesigned a robotic-arm joint bracket with an integrated bearing housing to reduce unnecessary material while maintaining its functional requirements; performed SolidWorks Simulation FEA using ABS material and 100 N loading per edge to evaluate structural performance.
- Performed static payload torque calculations for a 3-DOF robotic arm and selected servo motors based on the calculated joint torque requirements for a 1 kg payload.
- Designed a complete 5-DOF robotic arm assembly in SolidWorks, including the robotic links, joint structures, motor mounts, and overall mechanical architecture as a mechanical CAD concept.

EDUCATION

Bachelor of Science (B.Sc.) in Mechatronics Engineering

Jun 2021 - Jun 2026

Karabük University, Türkiye

CERTIFICATIONS & PROFESSIONAL MEMBERSHIPS

- Saudi Council of Engineers (SCE) - Engineer Membership (Active)
- Certified SOLIDWORKS Design Professional - Mechanical Design (CSWP), Dassault Systèmes | Aug 2026
- Certified SOLIDWORKS Associate — Mechanical Design (CSWA), Dassault Systèmes | May 2026
- ROS 2 Fundamentals, Karabük University | May 2025

LANGUAGES

Arabic - Native | English - Upper-intermediate | Turkish - Intermediate